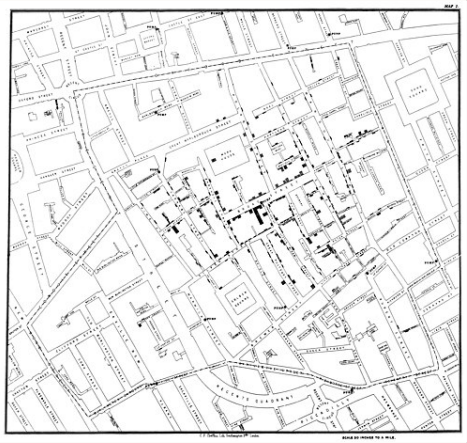
PUPIL NAME: _____ CLASS: _____ TEACHER: _____

YEAR 7 ENQUIRY • C.AD 43–PRESENT

Water & Sanitation Through Time

Conquering Squalor, Cholera & The Battle for Public Health



PRIMARY VISUAL EVIDENCE • EPIDEMIOLOGICAL CARTOGRAPHY — Dr John Snow (1854)

During the catastrophic 1854 Soho cholera epidemic, Dr John Snow mapped every recorded death as a black bar on a street plan, proving that water from the Broad Street pump—not airborne miasma—spread the lethal disease:

1. THE BROAD STREET PUMP EPICENTRE:
Clustered bars show over 500 fatal attacks occurred within a 250-yard radius of the Broad Street well pump.

2. THE BROAD STREET BREWERY ANOMALY:
Snow found none of the brewery's 70 workers contracted cholera because they drank free malt liquor instead of pump water.

3. LEAKING CESSPIT CONTAMINATION:
Subsequent excavation revealed a defective Victorian brick cesspit leaking raw sewage just three feet from the public well.

4. REMOVAL OF THE PUMP HANDLE:
Snow persuaded parish guardians to remove the pump handle on 8 September 1854, halting the localized Soho outbreak.

WATER VECTOR	SPATIAL DATA	CIVIC REFORM
Ingested pathogen replaces airborne miasma.	Spot mapping establishes modern epidemiology.	Scientific proof drives 1875 Public Health Act.

DISCIPLINARY SIGNIFICANCE: Snow's spot map shattered the centuries-old miasma theory, demonstrating that public health breakthroughs require empirical data and state intervention rather than moral blame.

DISCIPLINARY HINGE QUESTION: HISTORICAL EVALUATION

Did Dr John Snow's removal of the Broad Street pump handle prove the contagion was in the water, or did the miasma theory persist because Victorian authorities could not conceive of invisible living microbes before Pasteur?

Primary Visual Evidence: *On the Mode of Communication of Cholera* • Wellcome Collection Archive (1855)

WELLCOME ARCHIVE • ACCESSION EPID-1854-SNOW

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE	LEARNT	REVISED
Enquiry 1: Roman Britain & Fishbourne Palace (c.AD 43–410) *Gravity aqueducts, thermae, cloacae, and lead water pipes as instruments of imperial power.*	☐	☐
Enquiry 2: The Middle Ages & Monastic Conduits (c.1100–1500) *Monastic settling tanks vs urban squalor, the 1348 Black Death, and the 1388 Cambridge Statute.*	☐	☐
Enquiry 3: Early Modern Cesspits & Harington's WC (c.1500–1750) *The flush toilet invention (1596), the 1613 New River, Pepys's cellar crisis, and the 1665 Great Plague.*	☐	☐
Enquiry 4: Boomtown Squalor, Asiatic Cholera & Bazalgette's Sewers (1831–1865) *Industrial urbanization, Chadwick's 1842 Report, the 1858 Great Stink, and Bazalgette's intercepting sewers.*	☐	☐
Enquiry 5: Germ Theory, Dr Snow & The 1875 Revolution (1854–1900) *The 1854 Broad Street pump, Pasteur's Germ Theory, Koch's bacterium discovery, and the 1875 Public Health Act.*	☐	☐

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AUDIO • QUIZ

Was Roman water engineering built to protect ordinary people, or to show off imperial power and luxury?

1 AD 43 • How did the Roman invasion of AD 43 trigger an urban sanitation crisis?

- Dispersed Celtic Roundhouses
- Four Imperial Legions
- Cramped Walled Cities

2 c.AD 43–100 • How did Roman engineers use gravity to transport millions of gallons of water?

- Distant Mountain Springs
- Mathematical Downward Gradient
- Castella Water Towers

3 c.AD 100 • How did monumental public bathhouses promote cleanliness and imperial culture?

- Subterranean Hypocaust Heating
- Olive Oil & Bronze Strigils
- Frigidarium Cold Plunge

4 c.AD 150 • Why did forensic science expose severe flaws in Roman hygiene and health?

- Communal Sponge on a Stick
- Intestinal Parasite Eggs
- Toxic Lead Water Pipes

5 AD 410 • Why did Roman water engineering abruptly collapse after AD 410?

- Imperial Power & Vanity
- Vanishing Roman Legions
- Thousand-Year Sanitation Void

Roman lead pipes at Fishbourne carried spring water under gravity pressure to garden fountains.

Question 2: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why the Romans built sophisticated water supply and sanitation systems in Britain after AD 43. [12 marks]

Stimulus: **The need for fresh water and hygiene in densely populated towns like Londinium**

Displaying imperial power, Romanitas, and luxury at sites like Fishbourne Palace

(Support each factor with your own precise historical evidence)

1. URBAN DENSITY & HYGIENE	2. IMPERIAL POWER & ROMANITAS	3. MILITARY READINESS & VERDICT
Explain how founding crowded stone towns (Londinium, York) forced Romans to construct gravity aqueducts and flushing latrines to prevent devastating epidemics [1.1, 1.2].	Explain how grand thermae, heated hypocausts, and luxury lead pipes at Fishbourne Palace were designed to display imperial supremacy (*romanitas*) [1.3, 1.5].	Weigh whether practical military survival or imperial vanity was more decisive in driving Roman engineering [1.4, 1.5].

KEY TERMS: **Aqueduct • Cloaca • Thermae • Miasma • Cesspit**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

MIND-MAP MISSION Turn to **Page 8** and use the app's *Consolidate Station* to branch out the **c.AD 75 Roman Aqueducts & Fishbourne** node (aqueducts, lead pipes, thermae & latrines).

Was medieval public health squalor caused by a lack of money, or by mistaken medical beliefs like miasma?

AUDIO • QUIZ

1 1100–1250 • Why did the growth of medieval towns overwhelm traditional water supplies?

- Collapsed Roman Stone Aqueducts
- Booming Market Boroughs
- Unpaved Muddy Thoroughfares

2 c.1250–1348 • How did medieval towns attempt to manage human and animal waste?

- Overcrowded Timber Tenements
- Night-Time Gong Farmers
- Court Leet Fines & Rakers

3 c.1300 • Why did medieval monasteries enjoy far cleaner water than crowded towns?

- Settling Tanks & Lead Conduits
- Segregated Monastic Cloisters
- Titchfield Abbey Fishponds

4 1348–1350 • How did the 1348 Black Death challenge medieval understanding of disease?

- Yersinia Pestis in Fleas
- Poisonous Foul Air (Miasma)
- Forty-Day Port Quarantines

5 1388 • Did public health genuinely collapse, or have Victorians exaggerated medieval squalor?

- Statute of Cambridge (1388)
- Compulsory River Clean-Up
- Carole Rawcliffe Revisionism

Medieval monks drank spring water filtered through five settling tanks while townspeople drank from rivers.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why living conditions and public health in medieval towns were so much worse than in medieval monasteries. [12 marks]

Stimulus: **The design and funding of monastic water systems such as Canterbury Priory** | **The rapid, unregulated growth of commercial towns and lack of central taxation**

(Support each factor with your own precise historical evidence)

1. TOWN SQUALOR & LACK OF INFRASTRUCTURE	2. MONASTIC WEALTH & HYDRAULIC PLANNING	3. COMPARATIVE VERDICT
Explain how rapid population growth, unpaved streets, animal butchery, and leaking cesspits contaminated town water supplies [2.1, 2.2].	Explain how vast land endowments enabled monasteries (Canterbury, Titchfield) to construct settling tanks, lead conduits, and flushing reredorters [2.3].	Weigh whether financial wealth or geographical isolation was more decisive in explaining the health divide between towns and monasteries [2.3, 2.5].

KEY TERMS: **Aqueduct • Cloaca • Thermae • Miasma • Cesspit**

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MIND-MAP MISSION Turn to **Page 8** and use the app's *Consolidate Station* to branch out the **c.1165 Monastic Conduits** and **1388 Statute of Cambridge** nodes.

The 1388 Statute of Cambridge banned dumping animal dung and offal into rivers under heavy royal fines.

AUDIO • QUIZ

Was early modern filth caused by population explosion, or the lack of running water to make inventions work?

1 1500–1600 • How did London's rapid expansion create an unprecedented sanitation crisis?

- National Population Doubles
- 500,000 London Residents
- Cantilevered Timber Tenements

2 1596 • Why did Sir John Harington's revolutionary flush toilet fail to clean up Britain?

- Queen Elizabeth's Godson
- Valve-Operated Cistern
- Absence of Running Water Mains

3 1609–1613 • How did private enterprise attempt to bring clean spring water into London?

- Sir Hugh Myddelton
- 38-Mile Man-Made Canal
- Hollowed Elm Tree Trunk Pipes

4 1660–1665 • What does Samuel Pepys' diary reveal about domestic sanitation in Stuart London?

- Stepping into Neighbour's Waste
- Fractured Brick Cesspits
- Nightsoil Men & Filth Carts

5 1665 • Why did medical responses to the 1665 Great Plague remain fundamentally medieval?

- 100,000 Plague Fatalities
- Red Crosses & Watchmen
- Pounding Tobacco & Pomanders

Sir John Harington invented the flush toilet in 1596, but towns lacked running water mains to make it work.

Question 3: Evaluative Statement Essay [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

'Early Modern towns made substantial progress in water supply and sanitation between 1500 and 1700.' How far do you agree? [12 marks]

Stimulus: **Inventions and infrastructure like Harington's water closet (1596) and Myddelton's New River (1613)** | **Domestic conditions documented by Samuel Pepys and the devastating Great Plague of 1665** | *(Support each factor with your own precise historical evidence)*

1. TECHNOLOGICAL PROGRESS & INNOVATION	2. POPULATION PRESSURE & SQUALOR	3. SUSTAINED HISTORICAL VERDICT
Explain how Harington's flush toilet (1596) and Myddelton's New River (1613) demonstrated genuine engineering progress [3.2, 3.3].	Explain how explosive population growth, overflowing cesspits (Pepys), and the 1665 Great Plague proved ordinary conditions worsened [3.1, 3.4, 3.5].	Weigh whether private elite breakthroughs or widespread public squalor defined the era [3.5].

KEY TERMS: **Aqueduct • Cloaca • Thermae • Miasma • Cesspit**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

MIND-MAP MISSION Turn to **Page 8** and use the app's *Consolidate Station* to branch out the **1596–1660 Harington WC & New River Mains** node (valve flush, elm pipes, Pepys's cellar).

Samuel Pepys recorded stepping into his cellar in October 1660 to find his neighbour's cesspit had leaked through.

Roman Aqueducts, Monastic Conduits, Urban Cesspits & Harington

REVISION MIND-MAP CANVAS: At the end of each lesson, use the app's *Consolidate Station* to retrieve your keywords and branch out your mind-map across the open page (Problems & Beliefs to the left • Engineering & Laws to the right).

◂ GRAVITY SPRINGS & LEAD PIPES

c.AD 75 Roman Aqueducts & Fishbourne (Enq 1)

THERMAE & PUBLIC LATRINES ▸

◂ MONK WIBERT & PURE SPRINGS

c.1165 Canterbury Monastic Conduits (Enq 2)

FIVE SETTLING TANKS & PIPES ▸

◂ BUTCHERS' OFFAL & RIVER FILTH

1388 Urban Cesspits & Cambridge Act (Enq 2)

ROYAL FINES & GONGFERMORS ▸

◂ VALVE FLUSH WC & ELM PIPES

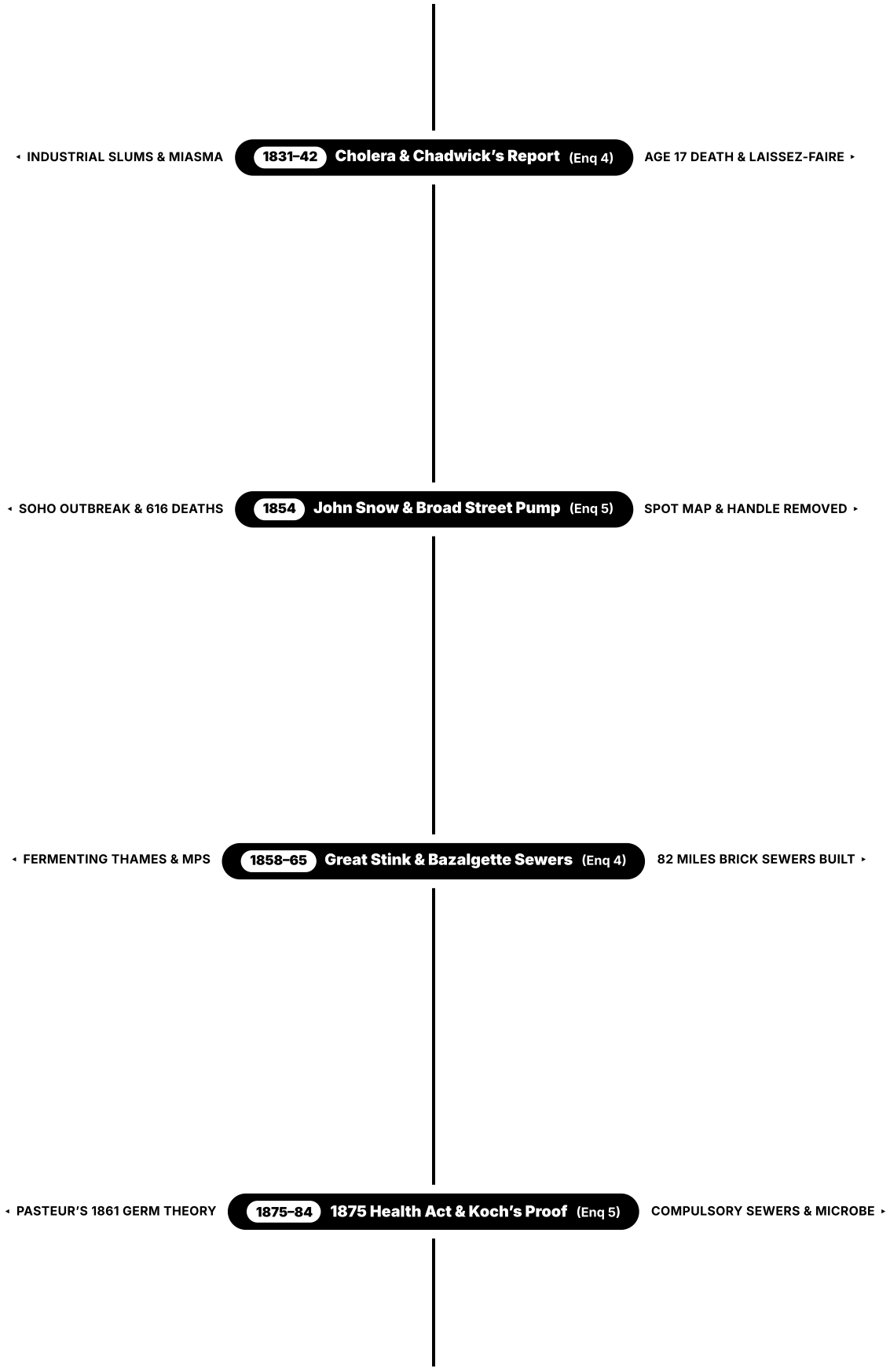
1596–1660 Harington WC & New River Mains (Enq 3)

PEPYS'S LEAKING CESSPITS ▸

CURRICULUM SPINE CONTINUES TO CHOLERA, SNOW & 1875 ACT (PAGE 9) ➔

Industrial Cholera, John Snow, Bazalgette & The 1875 Health Act

REVISION MIND-MAP CANVAS: At the end of each lesson, use the app's *Consolidate Station* to retrieve your keywords and branch out your mind-map across the open page (Outbreaks & Evidence to the left • Sewers, Science & Reform to the right).



WATER & SANITATION CURRICULUM SPINE COMPLETE (C.AD 75–1884) ➔

AUDIO • QUIZ

Was Victorian cholera conquered by civil engineering and fear of the Great Stink, or did politicians act out of genuine care for the poor?

1 1780–1840 • How did rapid industrial urbanisation create unprecedented slum squalor in British factory boomtowns?

- Steam-Powered Factory Boomtowns
- Unventilated Back-to-Back Slums
- Shared Overflowing Privy Middens

2 1831–1832 • Why did the arrival of Asiatic cholera in 1831 terrorize Victorian Britain and expose the failure of miasma theory?

- Sunderland Baltic Shipping Port
- 52,000 Dehydration Fatalities
- Tar Barrels & Miasma Fallacy

3 1842–1848 • Why did Edwin Chadwick's 1842 Report fail to clean up Britain despite proving that filth caused disease?

- Chadwick's Sanitary Report 1842
- Poverty Caused by Slum Filth
- Permissive 1848 Public Health Act

4 June 1858 • How did the environmental catastrophe of the 'Great Stink' finally shatter Victorian laissez-faire inertia?

- 200,000 Flushing Water Closets
- Fermenting Thames Sludge 35°C
- Parliament Panic & £3m Grant

5 1858–1865 • Why was Sir Joseph Bazalgette's intercepting sewer network considered the greatest engineering triumph of the Victorian age?

- 82 Miles Intercepting Sewers
- 318 Million Bricks & Portland Cement
- Gravity Gradient & Eastney Pumping

Edwin Chadwick proved in 1842 that industrial labourers in Manchester died at an average age of just 17.

Question 2: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why living conditions and public health in British boomtowns worsened dramatically during the Industrial Revolution. [12 marks]

Stimulus: **The rapid construction of unventilated back-to-back housing and shared privy middens** | **The arrival of Asiatic cholera in 1831 and the persistence of miasma theory**

(Support each factor with your own precise historical evidence)

1. INDUSTRIAL URBANISATION & SQUALOR	2. ASIATIC CHOLERA & MIASMA DOGMA	3. LAISSEZ-FAIRE INERTIA & VERDICT
Explain how steam-powered factories caused population explosion and flimsy back-to-back tenements with zero municipal drainage [4.1].	Explain how the 1831 Asiatic cholera epidemic struck and why belief in airborne miasma delayed clean water interventions [4.2, 4.3].	Explain how ratepayer opposition and laissez-faire politics blocked Chadwick's 1848 Public Health Act until the 1858 Great Stink [4.3, 4.4].

KEY TERMS: **Aqueduct • Cloaca • Thermae • Miasma • Cesspit**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

MIND-MAP MISSION Turn to **Page 9** and use the app's *Consolidate Station* to branch out the **1831–42 Chadwick Report** and **1858–65 Bazalgette Sewers** nodes.

AUDIO • QUIZ

Was cholera defeated by scientific observation, laboratory discovery of bacteria, or the introduction of compulsory government legislation?

1 1854 • Why did the medical establishment refuse to consider water as the cause of the 1854 Soho outbreak?

- Soho Broad Street Epicentre
- Over 500 Deaths in Ten Days
- Entrenched Miasma Medical Dogma

2 September 1854 • How did Dr John Snow use cartographic mapping and anomalous data to prove cholera was waterborne?

- Empirical Cholera Spot Map
- Broad Street Pump Clustering
- Lion Brewery Zero Fatalities

3 1854–1855 • What physical evidence finally proved how the Broad Street pump well had become contaminated?

- Removed Broad Street Pump Handle
- Infant Frances Lewis at No. 40
- Fractured Brickwork & Leaking Cesspool

4 1861–1884 • Why was laboratory bacteriological proof necessary to finally overthrow the ancient miasma theory?

- Pasteur's Swan-Neck Flask Experiments
- Germ Theory of Disease 1861
- Koch Isolates Vibrio Cholerae 1884

5 1875 • Why did the compulsory nature of the 1875 Public Health Act mark the definitive end of laissez-faire in Britain?

- Public Health Act 1875
- Compulsory Municipal Officers & Water
- End of Laissez-Faire Ideology

Michael Faraday warned in 1855 that the Thames was an opaque, fermenting sewer of human waste.

Question 2: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why public health in Britain transformed dramatically between 1854 and 1900. [12 marks]

Stimulus: **Dr John Snow's Soho investigation and spot map (1854)** | **Louis Pasteur's Germ Theory and the 1875 Public Health Act**

(Support each factor with your own precise historical evidence)

1. SCIENTIFIC EPIDEMIOLOGY (SNOW)
Explain how Dr John Snow used spatial mapping around the Broad Street pump to prove cholera was an ingested waterborne poison [5.1, 5.2, 5.3].

2. BACTERIOLOGY (PASTEUR & KOCH)
Explain how Pasteur's Germ Theory (1861) and Koch's isolation of Vibrio cholerae (1884) permanently destroyed miasma dogma [5.4].

3. COMPULSORY STATE REFORM (1875)
Explain how the compulsory 1875 Public Health Act ended laissez-faire and forced all councils to provide clean water and sewers [5.5].

KEY TERMS: **Aqueduct • Cloaca • Thermae • Miasma • Cesspit**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

MIND-MAP MISSION Turn to **Page 9** and use the app's *Consolidate Station* to branch out the **1854 Broad Street Pump** and **1875–84 Public Health Act & Koch** nodes.

KEY STAGE 3 HISTORY • PUPIL PROGRESS & ASSESSMENT RECORD

PUPIL NAME: _____ **CLASS:** _____ **TARGET KS3 LEVEL:** _____

KEY STAGE 3 HISTORY ASSESSMENT PATHWAYS • LEVEL DESCRIPTORS & CRITERIA — **YEARS 7–9 STANDARDS**

EMERGING (1–2)	EMERGING+ (3)	EXPECTED (4–5)	EXPECTED+ (6–7)	GREATER DEPTH (8–9)
Identifies key names, dates, and terms; describes what happened using simple statements and structured writing frames.	Describes reasons why events occurred; begins to explain causes and consequences supported by basic historical facts.	Constructs structured paragraphs explaining multiple causes and consequences; substantiates arguments with precise historical dates, key terms, and named figures.	Weighs competing factors and perspectives; explains why certain causes were more decisive than others using detailed, accurate evidence across the unit.	Crafts sustained, analytically rigorous arguments reaching a substantiated historical judgement; masterfully synthesizes evidence across the unit to evaluate complex turning points and long-term significance.

DATE	ENQUIRY • ASSESSMENT FOCUS	DO NOW (/5)	KS3 LEVEL / EFFORT	TEACHER COMMENT / SPELLINGS
	Enquiry 1: Roman Britain & Fishbourne Aqueducts, thermae & imperial hygiene infrastructure	/ 5		
	Enquiry 2: Middle Ages & Monastic Conduits Monastic settling tanks vs urban filth & 1388 Statute	/ 5		
	Enquiry 3: Early Modern Cesspits & Harington The 1596 WC, New River 1613 & Pepys's cellar overflow	/ 5		
	Enquiry 4: Boomtown Squalor & Bazalgette 1831 Asiatic cholera, Chadwick 1842 & Great Stink 1858	/ 5		
	Enquiry 5: Germ Theory, Snow & 1875 Revolution Broad Street pump 1854, Koch 1884 & 1875 Public Health Act	/ 5		
	Independent Study & Notes Portfolio Pages 14–15 Enquiry Notes, Reflection & Revision Tasks	—		
	November Assessment Masterclass Task 12-Mark Timed Disciplinary Assessment Paper	—		
★	**UNIT RETRIEVAL STARTER TOTAL:**	/ 25	Overall Retrieval Grade: [Emerging • Expected • Greater Depth]	

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE — **ONLINE RECALL**

ENQUIRY 1	ENQUIRY 2	ENQUIRY 3	ENQUIRY 4	ENQUIRY 5
Roman Britain	Middle Ages	Early Modern	Squalor & Sewers	Snow & 1875 Act
SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ
Best Score: [/ 8]	Best Score: [/ 8]	Best Score: [/ 8]	Best Score: [/ 8]	Best Score: [/ 8]

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